

An introduction to assembly language and other tool software I created out of necessity.

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An introduction to assembly language and other tool software that I created out of necessity.

This article introduces cross-assemblers and cross-disassemblers for the 6809.

[October 2nd Update] A revised version of the 6800/6809 disassembler was uploaded on October 2nd.

I mainly enjoy creating programs for the 6809, and I'd like to share some of the programs I've written out of necessity.

Basically, the process involves creating the source code on Windows, assembling it with a custom-made cross-assembler, and then transferring the generated binary to a real machine such as a 6809SBC for execution.

- (1) Cross-assembler, cross-disassembler
- (2) Transfer tool to the actual device
- (3) File format conversion tool

So, over the past 10 years or so, I've been creating cross-assemblers and cross-disassemblers for the 6800 and 6809. (I also created cross-assemblers for the 6801 and 6303 as needed.)

I've introduced some of the programs in previous blog posts, but I've made various modifications to them since then, so I'll introduce them all here.

The things I will introduce are

- (1) Assembler and disassembler

6809 cross-assembler 6809AsmWin.exe and cross-disassembler 6809DasmWin.exe

6800 cross-assembler 6800AsmWin.exe and cross-disassembler 6800DasmWin.exe

- (2) Transfer tool

The transfer tool I often use is virtual drive creation software. I've created four different versions to suit various DOS (or similar) systems.

FlexDrvWin.exe for Flex9

OS9DrvWin.exe for OS-9

FBasDrvWin.exe for F-BASIC3.0

CpmDrvWinFT.exe and CpmDrvWinRS.exe for CP/M-80

(3) Other tools and software

CvtDskD77.exe is a software used to convert the format of floppy disk images.

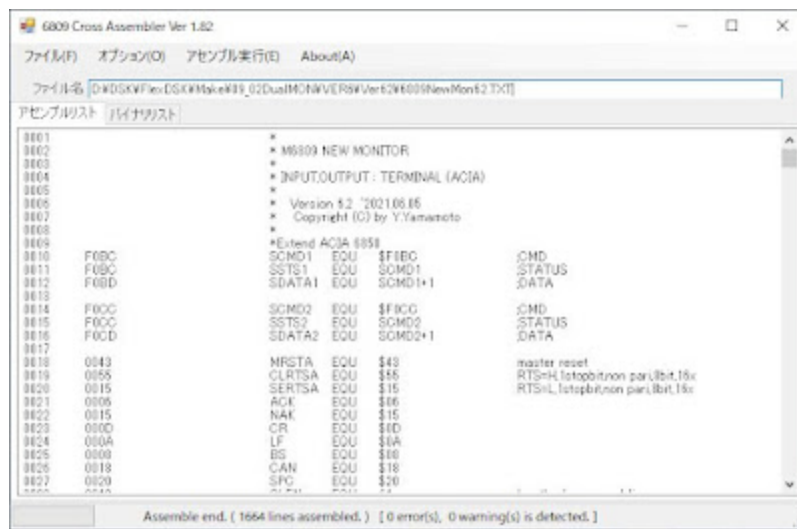
CvtMotHexBin.exe is a software program that converts binary files to Bin, Hex, and Mot formats.

Of the software listed above, all except CvtMotHexBin.exe were built using VC++2019.

I'll add a brief explanation of each software program.

(1) Cross-assembler

Simply select a file and press the "Execute Assemble" button. It also offers features such as displaying only errors and disabling warnings. The results can be saved in LST, BIN, MOT, and HEX formats. A notable feature is the local label function. Enabling this function allows you to declare commonly used labels, such as LOOP, in multiple locations.



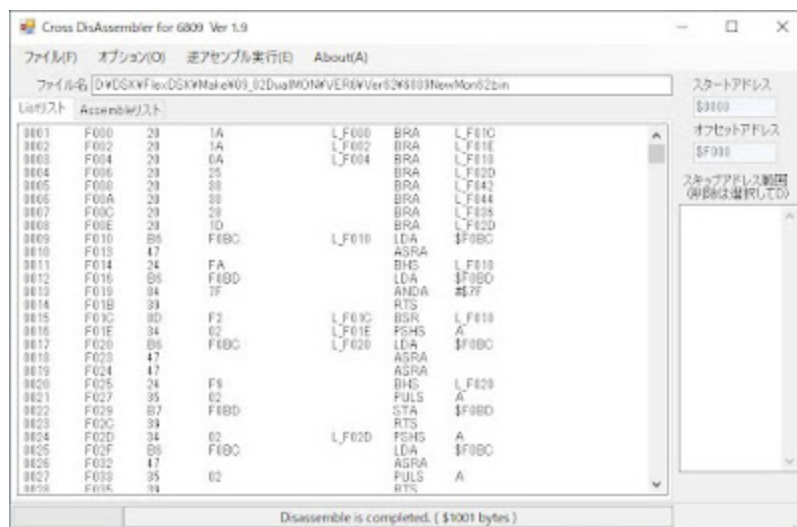
6809 Cross-Assembler

(2) Cross Disassembler

The program can read not only BIN files but also MOT and HEX files as source files. The "Start Address" on the far right is \$0000 for BIN files, but the starting address is displayed for MOT and HEX files. The "Offset Address" is used to move the starting address of the disassembled file. The starting address is the sum of the start address and the offset address. Labels take the form of an address preceded by "L_", such as L_1234.

A key feature is that by designating a region presumed to be a data area as a "skip address range," that range is treated as a pseudo-instruction called either FCB or FCC. Which it becomes depends on whether the first byte of the range is between \$20 and \$7F; otherwise, it becomes FCC. (This is a rather simplistic classification, though...)

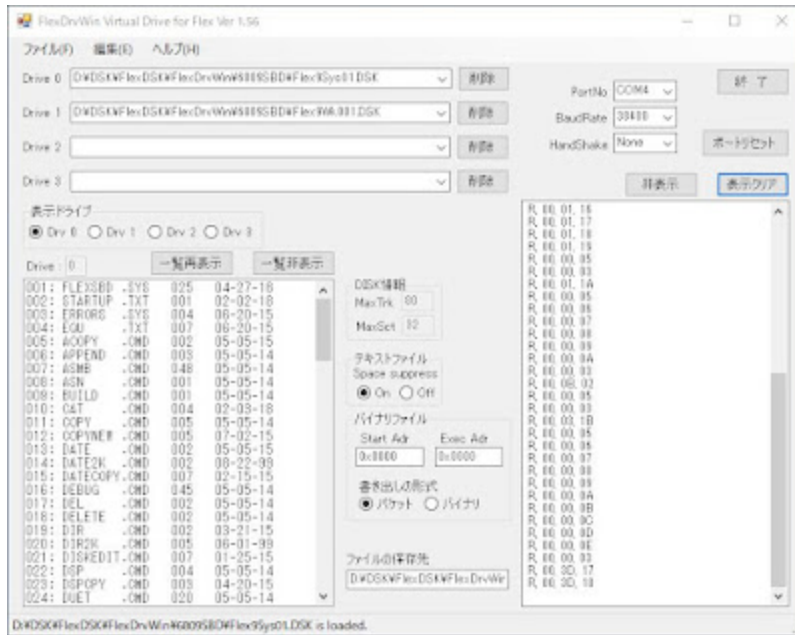
The supported file formats are ASM and LST, but ASM files can be directly processed by assembly language.



6809 Cross DisAssembler

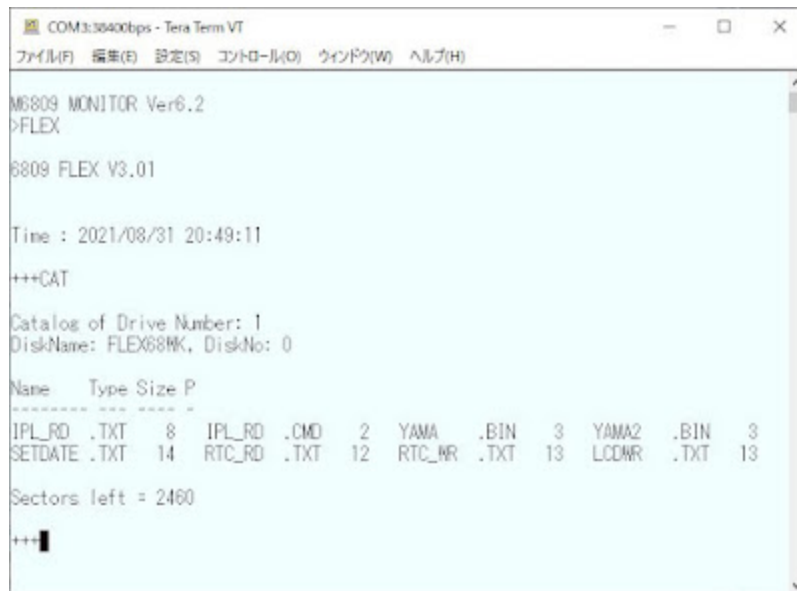
(3) Virtual drive creation software

The first version I created for Flex9 is shown below. If you set disk image files for drives 0-3, their contents will be displayed in the box in the lower left. Regarding image file formats, it can read not only D77 (D88) format but also DSK format (simply a sequential arrangement of sector data). Dragging and dropping files into this box will write them to the image file. You can select files in the box with the left mouse button and right-click to display a menu, allowing you to "Read," "Delete," and "Rename." To read all files at once, use the "Edit" -> "Read All Files" menu.



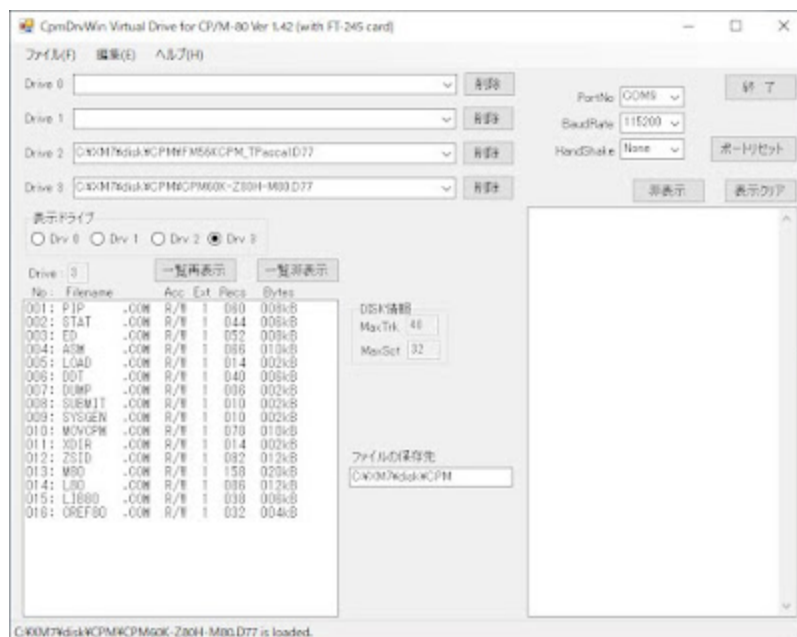
FlexDrvWin.exe

By setting up a communication path between the microcomputer/PC (such as the 6809SBC or FM-7), you can read and write data from the microcomputer/PC just like a regular drive. The diagram below shows an example of reading from Flex9 on a 6809SBC. Flex9 is started using the FLEX command on the monitor, and the CAT command of Flex9 is executed. The box on the right in the diagram above displays the commands sent from the SBC in R/W, Drv, Trk, and Sct formats.



Communication with 6809SBC

As an example of other software, the screen of CpmDrvWin.exe is shown below. Although the display and operation are all the same, internally, the processing is of course tailored to each FMS (File Management System).

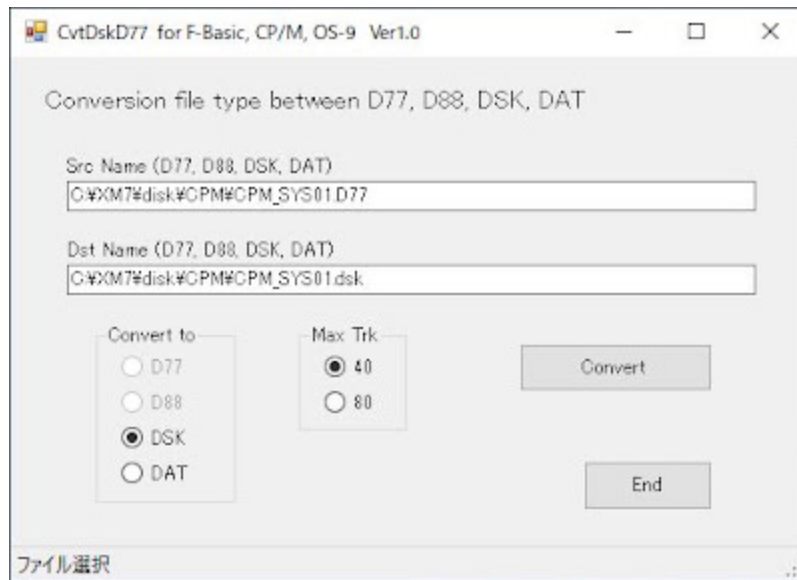


CpmDrvWin.exe

(4) File format conversion software

(4-1) Disk image file conversion software

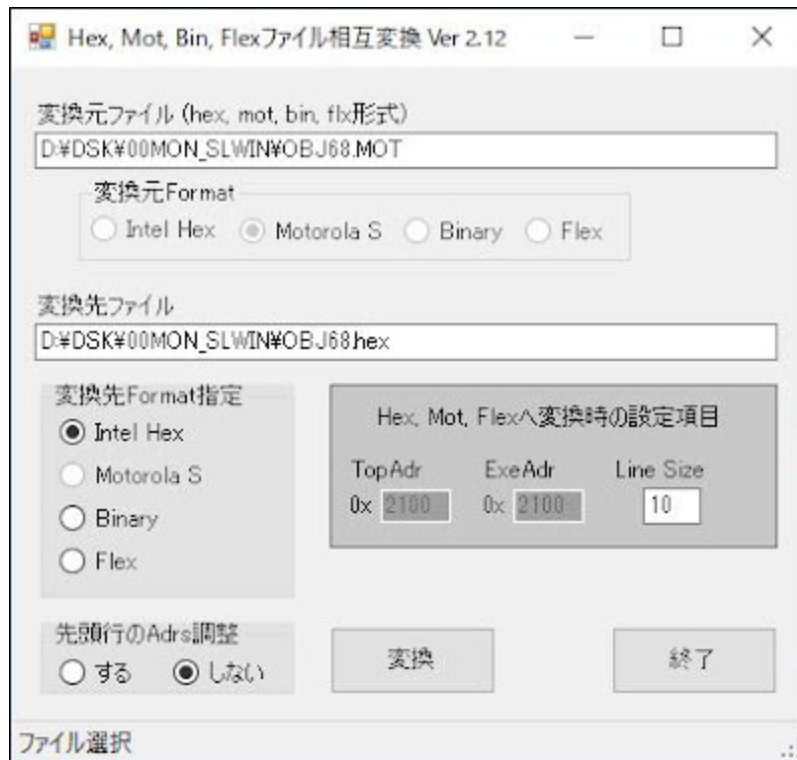
I usually use DSK files for disk image files because they are easier to read, but for XM7 and GOTEK, I need to convert them to D77 format. Until now, I've been using console software created for each DOS version, but that was inconvenient, so I created form software that works with all DOS versions. It automatically detects whether the number of tracks is 40 or 80, so it can be used with either.



D77, DSK, DAT conversion software

(4-2) Binary file conversion software

This software converts between Bin, Hex, and Mot files. It can also convert to Flex9 binary format.



Bin, Hex, Mot format conversion software

The software mentioned above was created out of necessity over the past 10 years or so. In particular, the assembler and disassembler were self-taught, created without any knowledge of standard structures, but they've been quite useful to me. (Occasionally, when I encounter unexpected behavior, I compare the results with those of other assemblers, but I believe the current version is fine.)

I have confirmed that Mikbug9, Mikbug20, TL1, and TinyBasic (Tokyo Denki University version) can be assembled for the 6800. In GAME III, there are many non-standard notations where a single character is enclosed in double quotes (") instead of single quotes ('), so I had to correct that ("A" → #'A').

For the 6809, I was able to assemble it by correcting a listing error in ASSIST09 (CPX → CMPX) and adding a single quote after the space in four places, changing #'(sp) to #'(sp)'.

I'll upload these to [OneDrive](#), so if you try using them and encounter any problems, please let me know.

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